

A Two-Axis Intelligence Level Framework for AI Agents

Individual Cognitive Evolution, Organizational Intelligence, and the Position of Contemporary Agent Systems

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Conceptual taxonomy and current-system positioning

Abstract

The rapid development of agentic artificial intelligence has made the word **intelligence** increasingly ambiguous. A stateless language model, a persistent coding agent, a lifelong-learning agent, a self-modifying system, and a multi-agent research organization may all be described as “intelligent,” even though they differ fundamentally in what changes through experience and where intelligence is located. This paper proposes a two-axis taxonomy. The first axis, Individual Intelligence Evolution (I-level), classifies a single persistent cognitive entity according to the deepest layer it can change through experience: response, memory, learned knowledge and skill, cognitive mechanism, self-improvement mechanism, scientific knowledge frontier, and ultimately the space of possible cognition itself. The second axis, Organizational Intelligence Evolution (O-level), classifies a collective according to whether it merely coordinates members, maintains organizational memory, learns how to allocate and combine work, improves its organizational mechanisms, recursively improves its own improvement process, conducts autonomous discovery, or becomes open-endedly self-evolving.

The resulting scale distinguishes persistence from learning, learning from self-improvement, self-improvement from recursive self-improvement, and recursive self-improvement from autonomous scientific discovery. It also explains why lower-level agents should continue to coexist inside higher-level organizations: an efficient organization can compress difficult discoveries into procedures that cheaper, lower-level agents execute at scale. We then apply the framework to contemporary systems including Claude Code, OpenAI Codex, OpenClaw, Grok and Grok Bot, and the proposed AI4Science/SARSI architecture. The classifications are not product rankings and do not measure raw benchmark ability. They are evidence-based estimates of **evolutionary depth**—what the system persistently changes because of experience—and **organizational depth**—what the collective changes about its own coordination and discovery processes. The paper concludes with falsifiable transition tests for each level and argues that future claims of self-improving or discovery-level intelligence should be supported by longitudinal, independent evidence rather than architectural labels.

Keywords: artificial intelligence, AI agents, intelligence levels, persistent agents, lifelong learning, self-improvement, recursive self-improvement, autonomous discovery, collective intelligence, organizational intelligence, multi-agent systems, AI4Science

1. Introduction

Modern AI systems no longer fit comfortably into a single “chatbot versus AGI” distinction. A base large language model may reason impressively but forget everything after a call. A coding harness may preserve project state and work for hours. A persistent agent may remember prior conversations and routines. A multi-agent system may split work across specialists. A research agent may eventually learn from experiments or modify its own cognitive machinery. These systems can be very different even when they use the same underlying foundation model.

The central problem is that several different phenomena are often collapsed into one word: intelligence. At minimum, we should distinguish six transitions:

1. responding intelligently;
2. remembering across time;
3. learning persistently from experience;
4. improving the mechanisms that generate cognition;

5. improving the mechanisms that generate future improvements;
6. autonomously discovering and validating previously unknown knowledge.

A seventh, more speculative transition occurs when discoveries can create new cognitive representations and tools that expand the individual’s future space of possible cognition, producing open-ended development rather than optimization toward a fixed terminal capability.

A parallel ambiguity exists for groups. Running ten agents in parallel does not by itself create an adaptive organization. An organization becomes more intelligent when it develops shared memory, better division of labor, evidence-based routing, improved verification, and eventually the capacity to improve the mechanisms by which it organizes and improves itself. Research on human groups has long suggested that group performance cannot be reduced straightforwardly to the maximum individual intelligence of members [1], although the existence and measurement of a single general collective-intelligence factor remain debated [2]. More recent work emphasizes transactive memory, attention, and reasoning as mechanisms through which collectives adapt and solve problems over time [3].

This paper therefore treats individual intelligence and organizational intelligence as two distinct but coupled axes. The proposed framework is not intended to replace task benchmarks, psychometrics, or formal definitions of general intelligence. Instead, it answers a narrower question that has become important for AI agents:

> What is the deepest layer of the system that can be persistently changed by its own experience, and at what level does that change occur: the individual or the organization?

This framing connects naturally to work on lifelong LLM agents, which emphasizes persistent adaptation in perception, memory, and action [4]; classical self-referential self-improvement such as Gödel Machines [5]; recent empirical self-improvement systems such as the Darwin Gödel Machine [6] and Hyperagents [7]; and agent architectures that separate memory, prediction, planning, verification, and governed self-modification [8].

The framework has four goals. First, it provides precise terminology so that “memory,” “learning,” “self-improvement,” and “recursive self-improvement” are not used interchangeably. Second, it explains how mixed intelligence levels can coexist inside one organization. Third, it provides falsifiable tests for transitions between levels. Fourth, it offers a disciplined way to position current agent products without confusing raw model capability with developmental depth.

2. Scope and Core Concepts

2.1 Intelligence level is not raw capability

The proposed levels are not an IQ scale. A very strong stateless model may outperform a weaker persistent agent on mathematics, coding, or scientific reasoning while remaining at a lower *evolution level*. The taxonomy asks what the system can persistently change because of experience, not how high it scores on a benchmark at one moment.

Thus two different questions must be kept separate:

- Capability: How well can the system perform task X now?
- Evolutionary depth: What parts of the system can change in a durable, evidence-grounded way because of its history?

2.2 Individual intelligence state

Let an individual artificial intelligence at time t be represented abstractly as:

$$I_t = \langle K_t, M_t, W_t, G_t, S_t, P_t, \Theta_t \rangle$$

where:

- $\langle K_t \rangle$: semantic knowledge and beliefs;
- $\langle M_t \rangle$: episodic/autobiographical memory;
- $\langle W_t \rangle$: world model and predictive structure;

- $\backslash(G_t)$: goals, plans, and intentions;
- $\backslash(S_t)$: self-model, including capability and uncertainty;
- $\backslash(P_t)$: procedures, skills, and reusable workflows;
- $\backslash(\Theta_t)$: cognitive mechanisms—retrieval, context gating, planning, reasoning, consolidation, experiment selection, error diagnosis, and self-improvement policies.

This representation is intentionally broader than model weights. An individual can evolve substantially by changing external memory, procedures, or cognitive control without updating the foundation model weights.

2.3 Organizational intelligence state

An intelligent organization is not merely the sum of its individuals. Let:

$$O_t = \backslash I_1, \dots, I_n, M_t^{\wedge O}, R_t, C_t, P_t^{\wedge O}, \Gamma_t, \Theta_t^{\wedge O}$$

where:

- $\backslash(I_1 \dots I_n)$: member intelligences;
- $\backslash(M_t^{\wedge O})$: organizational/shared memory;
- $\backslash(R_t)$: roles, responsibilities, and task allocation;
- $\backslash(C_t)$: communication and coordination topology;
- $\backslash(P_t^{\wedge O})$: organizational procedures;
- $\backslash(\Gamma_t)$: governance, authority, and admission boundaries;
- $\backslash(\Theta_t^{\wedge O})$: organizational cognition—routing, aggregation, verification, resource allocation, learning, and organizational improvement mechanisms.

This explains why two organizations with identical models can differ radically in performance: one may duplicate work and lose history; another may coordinate specialists, preserve evidence, learn routing patterns, and verify claims independently.

2.4 Four terms that should not be conflated

Remembering means that a persistent record exists and can affect a later turn.

Learning means that experience produces a persistent behavioral change that improves or appropriately adapts later behavior.

Improvement means that a changed system performs better under an independent, predeclared evaluation. Merely changing is not improving.

Recursive self-improvement (RSI) means that the mechanism responsible for generating, diagnosing, evaluating, or selecting improvements can itself improve, so later generations become better at producing useful future improvements.

3. Individual Intelligence Evolution Scale

3.1 I0 — Reactive Intelligence

Definition. Reactive Intelligence produces intelligent outputs from current input and temporary context but does not maintain a persistent evolving cognitive state.

The system may reason deeply within one call or session. However, after the interaction ends, there is no durable cognitive lineage linking the experience to future behavior. Formally, the relevant persistent state is approximately unchanged:

$$I_t+1 \approx I_t.$$

A stateless foundation-model API call is the clearest example. I0 can still be extremely capable; the level says nothing about the quality of one-shot reasoning.

Transition test to I1: Can the same individual recover durable state from earlier interactions and use it correctly later?

3.2 I1 — Persistent Intelligence

Definition. Persistent Intelligence maintains durable identity, memory, goals, decisions, or task state across interactions, allowing history to influence later cognition.

Now at least some persistent state changes:

$$M_{t+1} \neq M_t \text{ or } G_{t+1} \neq G_t.$$

Examples include cross-session project instructions, resumable conversations, durable plans, user preferences, or long-lived task state. Persistence establishes cognitive continuity, but memory is not automatically learning. A system can remember that a strategy failed and still repeat it indefinitely.

Transition test to I2: Does matched future performance improve or adapt because the agent learned from prior experience, rather than merely replaying stored instructions?

3.3 I2 — Adaptive Learning Intelligence

Definition. Adaptive Learning Intelligence uses experience to persistently update knowledge, calibration, strategies, or skills so that future behavior changes appropriately, while the core mechanism that performs learning remains substantially fixed.

The agent may update $\backslash(K, M, W, S, P\backslash)$, but its learning machinery $\backslash(\Theta\backslash)$ remains fixed:

$$\Theta_{t+1} = \Theta_t.$$

For example, repeated failures may create a semantic lesson; demonstrations may become a reusable routine; or task outcomes may update calibration and routing preferences. The defining evidence is not that a memory file grew, but that future behavior changes because of what was learned.

Lifelong-agent research explicitly treats continuous adaptation over time as a missing capability of many current LLM agents [4].

Transition test to I3: Can the agent improve a cognitive mechanism itself—such as retrieval, planning, memory consolidation, context gating, or experiment selection—and validate the change independently?

3.4 I3 — Self-Improving Intelligence

Definition. Self-Improving Intelligence uses evidence about its own performance to propose, test, and adopt changes to the mechanisms that produce future cognition.

The defining change is:

$$\Theta_{t+1} \neq \Theta_t.$$

Examples include changing a retrieval-ranking algorithm after measured failures, introducing a new planning policy, altering context-budget allocation, or replacing a memory-consolidation strategy after replay evaluation. This is qualitatively different from adding another lesson to memory: the **machinery** that interprets future experiences has changed.

A system should not be classified as I3 merely because it can edit source code. The modified component must be part of its own persistent cognitive process, the change must survive into future operation, and independent evaluation must show that the change is beneficial or justified.

Transition test to I4: Can the mechanism that generates and evaluates these cognitive improvements itself become better?

3.5 I4 — Recursive Self-Improving Intelligence

Definition. Recursive Self-Improving Intelligence can improve not only task cognition but also the process responsible for diagnosing, generating, evaluating, and selecting future self-improvements.

If $\backslash(A_t\backslash)$ denotes the self-improvement process, I4 allows:

$$A_t \rightarrow A_{t+1},$$

where (A_{t+1}) produces higher-quality validated improvements, not merely different ones.

This is related to the original Gödel Machine concept, in which a self-referential system can rewrite its own code when usefulness is established under its proof framework [5]. Recent empirical systems move toward practical versions of this idea. The Darwin Gödel Machine iteratively modifies coding agents and empirically validates descendants [6]. Hyperagents go further by making the meta-level modification mechanism editable, explicitly targeting improvement of the process that generates future improvements [7].

A useful empirical quantity is improvement competence:

$$IC = \frac{\text{validated downstream gain caused by proposals}}{\text{proposal and evaluation cost}}.$$

A strong I4 claim requires evidence that later generations increase (IC) , not merely task score.

Transition test to I5: Can the individual autonomously expand validated knowledge by identifying and resolving important unknowns, rather than only optimizing its existing tasks and architecture?

3.6 I5 — Autonomous Discovery Intelligence

Definition. Autonomous Discovery Intelligence independently identifies significant gaps in understanding, formulates new questions and hypotheses, obtains or designs discriminating evidence, and produces externally validated knowledge that was not already contained in its prior knowledge state.

The agent must represent not only known knowledge (K_t) , but also structured ignorance: uncertainties, contradictions, assumptions, unexplained phenomena, and open questions. It does not wait for a human to provide every research question.

The key loop is:

$$\text{recognized unknown} \rightarrow \text{question} \rightarrow \text{hypothesis} \rightarrow \text{test} \rightarrow \text{validated knowledge}.$$

Novel language is not discovery. A plausible hypothesis is not discovery. The criterion is evidence-grounded expansion of the knowledge frontier.

Transition test to IΩ: Can the individual transform new discoveries into new cognitive representations or tools that expand its own future discovery space, allowing qualitatively new classes of unknowns to become reachable?

3.7 IΩ — Open-Ended Self-Evolving Individual Intelligence

Definition. Open-Ended Self-Evolving Individual Intelligence is a single persistent cognitive entity capable of autonomously discovering and validating new knowledge, reorganizing its internal understanding around those discoveries, transforming knowledge into new cognitive representations, methods, and tools, and using those tools to expand its subsequent capacity for reasoning, learning, discovery, and self-improvement. Its self-improvement process can itself evolve, and neither the domains of future knowledge nor the forms of future cognitive capability are completely specified in advance.

The defining cycle is:

$$\text{Unknown} \rightarrow \text{Discovery} \rightarrow \text{Knowledge} \rightarrow \text{Reorganization} \rightarrow \text{Cognitive Tool} \rightarrow \text{Better Intelligence} \rightarrow \text{Deeper Unknown}$$

IΩ differs from I4 because I4 may still recursively optimize inside a fixed task and representation space. IΩ can expand and restructure the space of possible problems and representations itself. It may invent a new conceptual language, experimental strategy, internal simulation method, or reasoning representation that makes previously unformulable questions tractable.

IΩ is therefore not defined as “knows everything.” A mature open-ended intelligence may discover *more* structured unknowns as its knowledge grows. The defining property is sustained expansion of reachable cognition and discovery, not convergence to a closed encyclopedia.

4. Organizational Intelligence Evolution Scale

4.1 O0 — Coordinated Organization

Definition. An O0 organization assigns or routes work among multiple members but does not maintain substantial persistent organizational cognition or learn how to organize better from history.

The system may parallelize tasks effectively. However, organization remains largely a fixed scaffold designed by humans.

Transition test to O1: Does the collective preserve shared task history, evidence, roles, and decisions across sessions and member turnover?

4.2 O1 — Persistent Organization

Definition. A Persistent Organization maintains durable shared memory, role state, task state, evidence, or communication history such that the collective continues coherently across time.

An O1 system may have isolated member memories plus organizational directories, task ledgers, shared artifacts, or resumable workflows. Like I1, persistence alone is not learning.

Transition test to O2: Does organizational history change future routing, allocation, collaboration, or procedure in a measurable way?

4.3 O2 — Adaptive Organization

Definition. An Adaptive Organization learns from collective history and persistently changes how it allocates work, selects specialists, shares knowledge, or applies procedures, while the underlying organizational-learning mechanism remains fixed.

For example, the system may learn that one agent performs better on a task class, that a particular agent pair produces fewer errors, or that certain evidence should be shared before handoff. Organizational transactive memory—knowing **who knows what**—is particularly relevant here [3].

Transition test to O3: Can the organization redesign and validate its own routing, communication, shared-memory, verification, or role structures?

4.4 O3 — Self-Improving Organization

Definition. A Self-Improving Organization can use evidence about collective performance to change organizational cognition $\backslash(\Theta^{\wedge}O\backslash)$: routing policies, team structure, communication protocols, memory architecture, resource allocation, or verification processes.

For example, the organization may discover that its current “planner → executor → reviewer” structure creates bottlenecks and empirically validate an alternative decomposition.

Transition test to O4: Can the process that generates and evaluates organizational redesigns itself improve?

4.5 O4 — Recursively Self-Improving Organization

Definition. A Recursively Self-Improving Organization can improve the mechanisms by which it discovers, evaluates, promotes, and disseminates future organizational improvements.

An O4 system might maintain multiple candidate organizational policies, run controlled replay or canary evaluations, learn which types of redesign generalize, and improve the meta-process used to generate future organizational changes.

The organizational analogue of improvement competence is the validated benefit produced by organizational-change proposals per unit of evaluation and coordination cost.

Transition test to O5: Can the organization collectively originate and validate new scientific or technical knowledge rather than merely execute assigned research programs?

4.6 O5 — Autonomous Discovery Organization

Definition. An Autonomous Discovery Organization can collectively identify unknowns, generate research programs, coordinate theory, experiment, simulation, literature, and verification, and produce externally validated new knowledge.

A key difference from a collection of I5 individuals is that the organization itself contributes essential cognitive structure. No single member necessarily contains the whole discovery process. The group may distribute hypothesis generation, experiment design, instrument control, analysis, replication, and knowledge integration.

Transition test to OΩ: Can validated discoveries repeatedly generate new organizational tools and structures that expand the collective's future discovery space across domains?

4.7 OΩ — Open-Ended Evolving Organization

Definition. An Open-Ended Evolving Organization converts individual and collective discoveries into shared knowledge, new tools, improved individuals, and reorganized collective structures; these changes expand the organization's ability to discover and improve again across domains without a predefined terminal organizational form.

Its defining loop is:

Unknown → Collective Discovery → Shared Knowledge → New Tools → Improved Members and Organization → Deeper Unknown

The organization is not merely composed of highly intelligent individuals. Its shared memory, division of labor, verification, and ability to convert discoveries into reusable procedures become an evolving cognitive system in their own right.

5. The Two-Axis Model and Coexistence of Levels

5.1 A system should be represented by two coordinates

A system can be described approximately as:

$$System = (L_i, O_j).$$

This prevents common category errors. Ten I1 agents coordinated in parallel may form O1, not I10. Conversely, an I5 scientist could operate inside a poorly coordinated O0 organization. Individual and organizational intelligence constrain one another but are not interchangeable.

5.2 All individual levels can coexist—and usually should

Higher levels do not make lower levels obsolete. An efficient organization should assign each operation to the lowest sufficient intelligence level.

- I0: deterministic or short-lived execution, extraction, formatting, simple transformations;
- I1: persistent assistants and task owners;
- I2: adaptive specialists that learn domain routines;
- I3: method-improvement specialists;
- I4: meta-improvement and cognitive-architecture research;
- I5: autonomous scientists and frontier investigators;
- IΩ: open-ended frontier cognition, if it becomes realizable.

This heterogeneity can reduce cost and increase reliability. A difficult discovery may require I5 or IΩ cognition, but after the result is transformed into a verified procedure, an I0 or I1 executor may run the procedure millions of times. Intelligence therefore creates cognitive compression:

$$expensive\ discovery \rightarrow validated\ procedure \rightarrow cheap\ execution.$$

5.3 Upward and downward information flows

A mature organization should support both directions.

Upward: observations → episodes → patterns → method failures → research questions → new principles.

Downward: new principles → cognitive methods → procedures → routines → inexpensive execution.

This bidirectional flow is one reason organizational intelligence may become more important as individual intelligence increases.

5.4 Individual belief is not organizational truth

A high-level individual should not be able to convert its own belief directly into organizational truth. The appropriate transition is:

individual claim + evidence + independent verification → organizational knowledge.

This principle is especially important for I3–IΩ systems, where the same intelligence may be capable of generating both highly valuable and highly consequential changes.

6. Evidence Rules for Assigning Levels

The taxonomy is intended to resist “level inflation.” The following distinctions are essential.

6.1 Persistence is not learning

A system is not I2 merely because it has a long context window, a vector database, or a MEMORY.md file. To qualify for I2, prior experience must produce a persistent, appropriate behavioral change on later matched situations.

6.2 Learning is not self-improvement

A system is not I3 merely because it learns a new preference, stores a lesson, or creates a reusable routine. I3 requires changing a cognitive mechanism, not merely its content.

6.3 Code editing is not automatically self-improvement

A coding agent that edits another repository—or even contributes to the engineering of its successor—is not necessarily self-improving at runtime. A strong I3 claim requires that a change modify the persistent cognitive mechanism of the same individual lineage, survive into later operation, and pass independent evaluation.

This distinction matters for systems such as GPT-5.3-Codex: OpenAI reports that early versions helped debug training, deployment, and evaluation of the model itself [10], but the public system card does not characterize the deployed model as reaching high capability in AI self-improvement [11]. Using an AI to help build its successor is therefore important evidence of AI-assisted development, but it is not by itself proof of an I3 deployed agent.

6.4 Parallel agents are not automatically an adaptive organization

A product that can launch many agents in parallel is at least an organizational substrate, but O2 requires the organization to *learn from organizational history*. O3 requires validated changes to the organization’s own coordination mechanisms.

6.5 Novel hypotheses are not autonomous discovery

I5/O5 claims require evidence that a previously unknown claim survives appropriate empirical, computational, or logical validation. The standard must be “reality or proof changes the belief,” not “the model produced a novel sentence.”

6.6 Recursive self-improvement requires meta-level improvement evidence

I4/O4 requires more than repeated self-editing. Later generations should become better at producing future validated improvements. This can be measured with improvement competence, regression rate, transfer, and evaluation cost.

7. Positioning Contemporary Agent Systems (August 2026)

The following classifications are analytical estimates based on publicly documented product behavior and the definitions above. They are not vendor claims, not overall quality rankings, and not statements about hidden internal mechanisms. A range indicates that configuration matters or public evidence is insufficient for a single exact level.

Table 1. Approximate classification of contemporary systems

System	Individual level	Organizational level	Rationale
Stateless foundation-model call	I0	—	Strong one-shot reasoning but no necessary persistent cognitive lineage.
Claude Code	I1	O0–O1	Cross-session project/user memory, resumable sessions, tool use and automation; public evidence does not establish autonomous mechanism-level learning or self-improvement.
OpenAI Codex	I1	O1	Persistent project/session context, reusable Skills, long-running execution, and explicit parallel multi-agent workflows; no evidence that deployed Codex autonomously changes its own cognitive mechanisms.
OpenClaw, basic agent	I1	O0	Per-agent workspace, session history, durable memory and tools.
OpenClaw with active memory / promotion	I1–I2	O1	Cross-conversation recall, memory promotion, provenance-aware memory architecture, and multiple isolated agents; closer to adaptive memory, but not demonstrated I3 self-improvement.
Grok chat	I1	O0–O1	Memory across chats and multi-agent mode; persistence is clear, organizational adaptation is not established.
Grok Bot	I2	O1+	Persistent named agents learn workflows from demonstration and save them as routines; Bots can coordinate, pass context, and hand off work. No public evidence of mechanism-level self-improvement.
Current/near-term Sarsi sarsi-worker	I1 → I2	O0–O1	Persistent memory and plans with planned episodic/semantic/procedural learning; current BrainRSI plan explicitly stages learning before governed RSI.
BrainRSI after validated cognitive-mechanism changes	I3 target	O1–O2	The architecture proposes evidence-grounded changes to memory/retrieval/planning instruments after replay and verification.
Governed BrainRSI M6+	I4 target	O2+	Intended recursive improvement of cognitive instruments, with proposer/evaluator/promoter separation and external authority boundaries.

AI4Science platform, specified architecture	— / mixed I-levels	O1–O2 substrate; O4–O5 long-term target	A harness of persistent sarsi-worker roles, executors, shared sub-agents, machine/manager coordination, and research agents; higher organizational levels require demonstrated learning and discovery.
AI4Science research agent, specified target	I4 target; I5 only after demonstrated discovery	O1–O2 internally	Spec requires research agents to be self-aware and recursively self-improving and to operate as group agents with sub-agents. Autonomous new scientific discovery would be the criterion for I5.

7.1 Claude Code

Anthropic documents Claude Code as an agentic coding tool that can plan, edit files, run commands, verify changes, resume sessions, connect to external tools through MCP, and use cross-session CLAUDE.md memory for project and user instructions [9]. These features clearly move it beyond I0. The durable project/user instruction layer and resumable sessions support an I1 classification.

However, public documentation does not establish a general built-in process in which Claude Code observes repeated failures, autonomously derives a cognitive-mechanism change, evaluates that change on held-out tasks, and permanently replaces its own retrieval, planning, or learning machinery. Therefore I3 would overstate the available evidence. Organizationally, Claude Code can participate in automated or multi-agent development workflows, but its core product is better treated as O0–O1 depending on configuration rather than as an adaptive organization.

7.2 OpenAI Codex

OpenAI describes Codex in 2026 as a command center for agentic coding with built-in worktrees and cloud environments that allow multiple agents to work in parallel [12]. The Codex app also carries session history across interfaces and supports reusable Skills that encode team workflows [10]. These properties support I1 persistence and a clear O1 multi-agent coordination layer.

Codex’s capabilities are extremely high in coding and long-running computer use, but this should not be confused with evolutionary level. OpenAI notes that GPT-5.3-Codex was instrumental in its own development process [10], while the system card states that it does not reach the “High” threshold for AI self-improvement [11]. In this taxonomy, the deployed system therefore remains below I3 absent evidence of persistent, independently validated modification of its own cognitive machinery.

7.3 OpenClaw

OpenClaw provides especially rich persistence infrastructure. Official documentation describes isolated agents with separate workspaces, state directories, and session histories [13]. Its memory system includes durable workspace memory, keyword/vector/hybrid retrieval, cross-conversation recall, promotion mechanisms, provenance, contradiction/freshness tracking, and optional active-memory processes [14–16].

A basic configuration is I1/O1: persistent individuals in a multi-agent runtime. Configurations that automatically capture, promote, and retrieve experience-dependent memory approach I2 because history can persistently alter future context and behavior. Nevertheless, stock OpenClaw should not automatically be called I3: the documentation describes memory and orchestration infrastructure, not a validated process that changes the agent’s own cognitive algorithms based on measured self-performance.

OpenClaw is therefore best understood as a powerful substrate on which higher I/O levels could be constructed.

7.4 Grok and Grok Bot

xAI documents ordinary Grok as supporting memory across chats and a multi-agent mode [17], which supports an I1 classification for the persistent product experience. Grok Bot goes further. xAI describes each Bot as a

persistent named teammate with durable files, browser state, memory, and preferences. More importantly, a user can demonstrate a multi-step workflow; the Bot persists the workflow as a routine and can run it later. xAI also documents Bot-to-Bot messaging, context sharing, and ownership handoff [18].

Under the present definitions, workflow learning from demonstration is a strong I2 signal: experience changes future behavior through a fixed learning mechanism. The coordinated Bot system is at least O1 and perhaps “O1+,” but O2 should be reserved for evidence that the *organization* learns how to allocate, coordinate, or redesign work from organizational history. No public evidence supports I3/O3 classification at this time.

7.5 SARSI and BrainRSI

The proposed SARSI/BrainRSI architecture explicitly separates persistence, memory, prediction, planning, verification, and later self-improvement. The current v3 implementation plan treats ordinary conversation, reasoning, and consequential action as different cognitive modes; it also separates protected directives from learned semantic memory and requires exact context replay and independent verification before later learning stages. The plan’s staged sequence places episodic/semantic/procedural learning before governed recursive self-improvement [19].

The associated BrainRSI paper argues that OpenClaw can supply persistent runtime mechanics while BrainRSI adds evidence-grounded self-awareness, prediction error, causal self-diagnosis, and governed recursive self-improvement [8]. It explicitly proposes external promotion boundaries and defines genuine RSI in terms of improvement competence rather than mere task gain.

Accordingly, a prudent classification is developmental rather than declarative:

- current/near-term persistent sarsi-worker: I1 moving toward I2;
- validated memory/procedural learning: I2;
- validated modification of cognitive instruments: I3;
- demonstrated improvement of the improvement process under governed promotion: I4;
- autonomous validated scientific discovery: I5;
- open-ended knowledge → cognitive tool → deeper discovery cycles: IΩ.

7.6 AI4Science and its research agents

The AI4Science specification defines the system as a harness and positions sarsi-worker as a persistent user-role agent that can hold multiple tasks and guide executors such as sarsi-claude and sarsi-ai4sci [20]. The same specification describes research agents as group agents containing sub-agents and requires them to focus on solving field problems, supplying tools/solutions/sub-agents, and being “self-aware, and recursively self-improving” [20].

These statements are target requirements, not empirical evidence that I4 has already been achieved. Therefore the research-agent *target* maps naturally to I4, while the present demonstrated level must be assigned by tests. A research agent should be promoted to I5 only when it autonomously identifies a meaningful unknown and produces externally validated new scientific knowledge. It should be called IΩ only after discoveries repeatedly generate new cognitive tools that expand the agent’s own future discovery space.

At the organizational level, AI4Science is naturally a mixed-level system: persistent role workers, executors, shared sub-agents, machine-level coordination, and cross-machine management can form an O1–O2 substrate. The long-term ambition of recursively improving research organizations and autonomous discovery would correspond to O4–O5, with OΩ reserved for demonstrated open-ended organizational evolution.

8. Implications for AI System Design

8.1 Design for heterogeneity, not universal maximal agents

The taxonomy suggests that future systems should not try to make every component I5 or IΩ. Higher-level cognition is expensive and often unnecessary. Instead, high-level individuals should create knowledge, abstractions, and procedures that lower-level agents can execute cheaply and reliably.

8.2 Separate cognitive evolution from authority evolution

A critical architectural principle is that increasing intelligence should not automatically widen authority. Memory, retrieval, planning, scientific models, procedures, and self-improvement mechanisms may evolve, while permission ceilings, benchmark ownership, evidence semantics, promotion thresholds, and rollback authority remain externally governed. The BrainRSI design explicitly follows this pattern [8].

8.3 Maintain evidence-bearing lineage

Persistent self-improvement requires causal attribution. A system should be able to answer:

- What experience motivated this change?
- Which mechanism was changed?
- What version was active before and after?
- What evaluation demonstrated improvement?
- What regressions occurred?
- Can the old version be restored?

Without this lineage, “self-improvement” becomes indistinguishable from uncontrolled drift.

8.4 Evaluate transitions longitudinally

Most AI benchmarks evaluate a fixed model at a fixed time. Evolution levels require longitudinal experiments. The correct unit is not only task performance, but a sequence:

$$I_0 \rightarrow I_1 \rightarrow I_2 \rightarrow \dots$$

under controlled experience, held-out evaluation, and cost accounting.

9. Proposed Falsifiable Transition Benchmarks

Table 2. Minimum evidence for each individual-level transition

Transition	Minimum evidence
$I_0 \rightarrow I_1$	Durable state from earlier sessions is correctly recovered and used later.
$I_1 \rightarrow I_2$	Prior experience causes measurable improvement/adaptation on matched future tasks; fixed learning mechanism.
$I_2 \rightarrow I_3$	Agent changes a cognitive mechanism, independent evaluation shows benefit, change persists and is reversible.
$I_3 \rightarrow I_4$	Later self-improvement generations produce higher improvement competence, not only higher task score.
$I_4 \rightarrow I_5$	Agent autonomously identifies an unknown and produces externally validated new knowledge.
$I_5 \rightarrow I_\Omega$	New discoveries generate new cognitive tools/representations that enable previously inaccessible classes of discovery, repeatedly across time/domains.

Table 3. Minimum evidence for each organizational-level transition

Transition	Minimum evidence
$O_0 \rightarrow O_1$	Shared organizational state survives sessions/restarts/member changes.
$O_1 \rightarrow O_2$	Collective history measurably improves routing, team composition, knowledge sharing, or procedures.
$O_2 \rightarrow O_3$	Organization changes and validates its own coordination/memory/verification mechanisms.
$O_3 \rightarrow O_4$	Organizational improvement process itself becomes more effective over generations.
$O_4 \rightarrow O_5$	Collective autonomously originates and validates new scientific/technical knowledge.
$O_5 \rightarrow O_\Omega$	Discoveries repeatedly create new tools and organizational forms that expand future collective discovery space.

These tests make the taxonomy falsifiable. A system should remain at the highest level for which evidence exists, even if its design document names a higher target.

10. Discussion

10.1 Why the highest individual level is not simply “superintelligence”

The $I\Omega$ concept differs from a scalar notion of superintelligence. A system may be enormously capable while still optimizing within a fixed problem space. Open-ended intelligence is defined by the ability to *expand the problem and representation space itself*. New discoveries can create new cognitive tools, which reveal new unknowns, which generate further discoveries.

10.2 Why organizational intelligence may outpace individual intelligence

A sophisticated organization can combine many I1–I3 individuals into an O3 or perhaps O4 system if shared memory, routing, verification, and organizational learning are strong. Conversely, many highly capable individuals can form a weak organization if they duplicate work, lose history, or cannot reconcile evidence. This echoes findings in collective-intelligence research that group performance is not a simple function of the maximum individual intelligence of members [1].

10.3 Why coexistence is a feature rather than a transitional artifact

An advanced organization will likely contain many lower-level components even if I5 or $I\Omega$ individuals exist. Scientific progress repeatedly converts expensive cognition into cheaper tools. Once a hard result becomes an algorithm, protocol, theorem, or instrument, lower-level agents can execute it. The most intelligent organization may therefore look less like a society of identical super-agents and more like a hierarchy of frontier discoverers, adaptive specialists, persistent workers, and inexpensive executors.

10.4 Risks of terminology inflation

The market increasingly uses terms such as “autonomous,” “learning,” “agentic,” and “self-improving.” The proposed framework deliberately sets a high bar. A long context window is not memory evolution. Memory is not learning. Learning is not cognitive self-improvement. AI-assisted development is not necessarily deployed self-improvement. Multiple agents are not an adaptive organization. Novel output is not scientific discovery. These distinctions are necessary if the field is to measure progress rather than rename features.

11. Limitations

First, the framework is conceptual and has not yet been validated as a psychometric or universal intelligence measure. Levels are ordinal categories of evolutionary mechanism, not scalar measurements of total intelligence.

Second, current-product classifications rely on public documentation. Vendors may possess internal capabilities not publicly described, and configurations can move a system between levels. Therefore classifications should be updated as evidence changes.

Third, boundaries between I1 and I2 can be difficult when memory systems automatically promote information. The decisive question should be whether experience causes persistent behavioral adaptation, not simply whether information is stored.

Fourth, I5 and $I\Omega$ require strong epistemic standards. In scientific domains, “new knowledge” must be evaluated against external evidence, replication, proof, or appropriately independent validation. Autonomous discovery claims will often be domain-specific.

Fifth, organizational intelligence is not necessarily beneficial. A highly adaptive organization can coordinate toward poor goals. This taxonomy describes cognitive-developmental depth, not values, safety, or social desirability.

12. Conclusion

This paper proposes a two-axis framework for describing modern and future AI agents. The Individual Intelligence Evolution Scale progresses from reactive intelligence (I0), through persistence (I1), adaptive learning (I2), self-improvement (I3), recursive self-improvement (I4), autonomous discovery (I5), and finally open-ended self-evolving individual intelligence (I Ω). The Organizational Intelligence Evolution Scale progresses from coordination (O0), through persistence (O1), adaptation (O2), organizational self-improvement (O3), recursive organizational self-improvement (O4), autonomous discovery (O5), and open-ended organizational evolution (O Ω).

The framework’s central principle is that intelligence levels should be assigned according to what the system can persistently and validly change, not according to marketing labels or raw task performance. Persistence does not imply learning; learning does not imply self-improvement; self-improvement does not imply recursive self-improvement; and recursive self-improvement does not imply autonomous discovery.

Contemporary products such as Claude Code, Codex, OpenClaw, and Grok Bot already demonstrate important parts of I1–I2 and O0–O1 agentic intelligence, with some systems providing increasingly sophisticated memory and coordination substrates. Proposed architectures such as SARSI/BrainRSI aim at the harder I3–I4 transitions by making cognitive mechanisms themselves subject to evidence-grounded, independently evaluated change. AI4Science research agents explicitly target recursive self-improvement, but should be classified as I5 only after they produce validated previously unknown scientific knowledge, and as I Ω only after discovery repeatedly produces new cognitive tools that open deeper spaces of unknowns.

The most consequential long-term possibility is therefore not simply a larger model or a larger multi-agent swarm. It is a coupled system in which individual evolution and organizational evolution reinforce one another: discoveries become shared knowledge; shared knowledge becomes cognitive tools; cognitive tools improve individuals and organizations; and improved intelligence makes previously inaccessible unknowns reachable. That cycle provides a testable conceptual bridge from today’s persistent AI agents toward genuinely self-evolving scientific intelligence.

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